Q14: Project Suggestions

DLS 2026 — Kukla

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1. **Linear probes (medium, 3–5 days).** Train linear probes to decode attributes (sentiment, part-of-speech, factual properties) from hidden states across layers; show where each becomes linearly decodable.

2. **Toy superposition (medium, 1 week).** Reproduce a toy model of superposition: train a small autoencoder on sparse features in a low-dim bottleneck and visualize the near-orthogonal feature directions emerging.

3. **Fact localization (larger, 1–2 weeks).** Use causal tracing on a small open model to localize a factual association, then attempt a targeted edit and measure specificity (does it change the target fact without breaking neighbors?).